

improper integral $\left\{ \begin{array}{l} 2 \text{ types} \\ \text{comparison theorem} \end{array} \right.$

MATH 126

A4 past final worksheet

1. (Fall 2024, #3) Determine whether the following integrals are convergent or divergent. Justify your answer.

$$(a) \int_0^1 \frac{1+e^{-x}}{x^3} dx$$

$$\geq \int_0^1 \frac{1}{x^3} dx$$

$$= \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x^3} dx$$

$$= \lim_{t \rightarrow 0^+} \left. -\frac{1}{2} x^{-2} \right|_t^1$$

$$= \lim_{t \rightarrow 0^+} -\frac{1}{2} + \frac{1}{2} t^{-2} = \infty.$$

$\Rightarrow$ Divergent!

$$(b) \int_1^{+\infty} \frac{2x}{x^3+1} dx$$

$$\leq \int_1^{\infty} \frac{2x}{x^3} dx$$

$$= 2 \int_1^{\infty} \frac{1}{x^2} dx$$

$$= 2 \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx$$

$$= 2 \lim_{t \rightarrow \infty} \left. -\frac{1}{x} \right|_1^t$$

$$= 2 \lim_{t \rightarrow \infty} -\frac{1}{t} + 1$$

$$= 2 < \infty. \quad \Rightarrow \text{Convergent!}$$

2. (Spring 2024, #2) Find the following antiderivatives.

(a) $\int \frac{\sqrt{4-x^2}}{x} dx$

$$x = 4 \sin \theta$$

$$\Rightarrow \cos \theta = \sqrt{1 - \left(\frac{x}{4}\right)^2}$$

Let $x = 4 \sin \theta$ $dx = 4 \cos \theta d\theta$

$$\int \frac{2 \cos \theta}{4 \sin \theta} 4 \cos \theta d\theta$$

$$= 2 \int \frac{\cos^2 \theta}{\sin \theta} d\theta$$

$$= 2 \int \frac{\cos^2 \theta \sin \theta}{\sin^2 \theta} d\theta$$

$$= 2 \left(- \int \frac{\cos^2 \theta}{1 - \cos^2 \theta} d \cos \theta \right)$$

(b) $\int \frac{x+1}{x^3+x} dx$

$$\cos \theta = u$$

$$= -2 \int \frac{u^2}{1-u^2} du$$

$$= -2 \int -1 + \frac{1}{1-u^2} du$$

$$= -2 \left[-u + \int \frac{1}{1-u^2} du \right]$$

$$= -2 \left[-u + \frac{1}{2} \int \frac{1}{1-u} + \frac{1}{1+u} du \right]$$

$$= 2u - \int \frac{1}{1-u} du - \int \frac{1}{1+u} du$$

$$= 2u + \ln(1-u) - \ln(1+u) + C$$

$$= 2 \cos \theta + \ln \left(\frac{1 - \cos \theta}{1 + \cos \theta} \right) + C$$

$$= 2 \sqrt{1 - \left(\frac{x}{4}\right)^2} + \ln \left(\frac{1 - \sqrt{1 - \left(\frac{x}{4}\right)^2}}{1 + \sqrt{1 - \left(\frac{x}{4}\right)^2}} \right) + C$$

(b)

$$= \int \frac{x+1}{x(x^2+1)} dx$$

$$= \int \frac{a}{x} + \frac{bx+c}{x^2+1} dx$$

$$= \int \frac{a(x^2+1) + bx^2 + cx}{x(x^2+1)} dx$$

$$= \int \frac{(a+b)x^2 + cx + a}{x(x^2+1)} dx$$

$$\Rightarrow a+b=0, C=1, a=1$$

$$\Rightarrow C=1, a=1, b=-1$$

Thus,

$$(b) = \int \frac{1}{x} + \frac{-x+1}{x^2+1} dx$$

$$= \int \frac{1}{x} + \frac{-x}{x^2+1} + \frac{1}{x^2+1} dx$$

$$= \ln x - \frac{1}{2} \ln(1+x^2) + \arctan(x) + C$$

3. (Spring 2024, #3) Consider the improper integral

$$\int_0^1 \frac{5 - \sin x}{x^{1/3} + x^3} dx.$$

A student, investigating this integral, makes the following claims:

I. $\frac{5 - \sin x}{x^{1/3} + x^3} \leq \frac{6}{x^3}$ when $0 < x \leq 1$

II. $\int_0^1 \frac{6}{x^3} dx$ is convergent.

III. It follows from Claims I and II that the integral is convergent.

(a) Identify the flaw in the student's argument. Circle the best response.

i. Claim I is false.

ii. Claim I is true but claim II is false.

iii. Claims I and II are true but it does not follow from them that the integral is convergent.

(b) Is the integral convergent or divergent? Carefully justify your answer.

(a) Claim I is true!

Claim II is false ($x=0$ singular point) $\Rightarrow$ Divergent.

(b), $\frac{5 - \sin x}{x^{1/3} + x^3} \leq \frac{6}{x^{1/3}}$

$$\begin{aligned} \int_0^1 \frac{6}{x^{1/3}} dx &= 6 \cdot \frac{3}{2} x^{2/3} \Big|_0^1 \\ &= 6 \cdot \frac{3}{2} = 9 < \infty \end{aligned}$$

$\Rightarrow$ Convergent!

4. (Spring 2023, #5) Evaluate the following improper integral:

$$\int_5^{\infty} \frac{3}{x^2 + x} dx$$

Hint: the integral is known to converge.

$$\int_5^{\infty} \frac{3}{x(x+1)} dx$$

$$= \int_5^{\infty} 3 \left(\frac{1}{x} - \frac{1}{x+1} \right) dx.$$

$$= \lim_{t \rightarrow \infty} 3 \int_5^t \left(\frac{1}{x} - \frac{1}{x+1} \right) dx$$

$$= \lim_{t \rightarrow \infty} 3 \cdot \left[\ln x - \ln(x+1) \right] \Big|_5^t$$

$$= \lim_{t \rightarrow \infty} 3 \left[\ln \frac{t}{t+1} - \ln \frac{5}{6} \right].$$

Note that as $t \rightarrow \infty$ $\lim_{t \rightarrow \infty} \frac{t}{t+1} = 1$.

Then

$$= 3 \left(\overset{=0}{\ln 1} - \ln \frac{5}{6} \right)$$

$$= -3 \ln \frac{5}{6}.$$

5. (Fall 2021, #2) Compute the following integrals. If an integral is determined to be improper, you must also justify its convergence.

(a) $\int_0^1 \frac{x}{\sqrt{1-(x-1)^2}} dx$ NOT improper!

$$= \int_0^1 \frac{x-1+1}{\sqrt{1-(x-1)^2}} dx = \int_0^1 \frac{x}{\sqrt{(1+x-1)(1-x+1)}} dx$$

$$= \int_0^1 \frac{x-1}{\sqrt{1-(x-1)^2}} dx + \int_0^1 \frac{1}{\sqrt{1-(x-1)^2}} dx = \int_0^1 \frac{x}{\sqrt{x} \sqrt{2-x}} dx$$

$$= \int_0^1 \frac{\sqrt{x}}{\sqrt{2-x}} dx$$

$$= \frac{1}{2} \int_0^1 \frac{1}{\sqrt{1-(x-1)^2}} d(x-1)^2 + \sin^{-1}(x-1) \Big|_0^1$$

$$\stackrel{u=(x-1)^2}{=} \frac{1}{2} \int_1^0 \frac{1}{\sqrt{1-u}} du + (\sin^{-1}(0) - \sin^{-1}(-1))$$

$$= -\frac{1}{2} \int_0^1 \frac{1}{\sqrt{1-u}} du + 0 - \frac{\pi}{2} = -\frac{1}{2} \int_0^1 \frac{1}{\sqrt{1-u}} du + 0 + \frac{\pi}{2}$$

(b) $\int_0^1 x^{1/4} \ln x dx$

$$= \sqrt{1-u} \Big|_0^1 + 0 + \frac{\pi}{2}$$

$$= 0 - 1 + 0 + \frac{\pi}{2} = \boxed{\frac{\pi}{2} - 1}$$

(b) Note that

$$\lim_{x \rightarrow 0} x^{1/4} \ln x$$

$$= \lim_{x \rightarrow 0} \frac{\ln x}{x^{-1/4}} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{-\frac{1}{x}}{-\frac{1}{4} x^{-5/4}} = 4 \cdot x^{1/4} = 0$$

NOT improper! Integration by parts

$$\int_0^1 x^{1/4} \ln x dx = \int_0^1 \frac{4}{5} \ln x d x^{5/4}$$

$$= \underbrace{\frac{4}{5} \ln x \cdot x^{5/4}}_{=0} \Big|_0^1 - \int_0^1 \frac{4}{5} \cdot \frac{1}{x} \cdot x^{5/4} dx$$

$$= - \int_0^1 \frac{4}{5} x^{1/4} dx = - \frac{4}{5} \cdot \frac{4}{5} x^{5/4} \Big|_0^1$$

$$= - \frac{16}{25}$$